Personal Statement

Zhenghao Chen · PhD Application — Graph Intelligence & Urban Resilience

My path into research began with a lifelong curiosity about geography. As a high-school student drawn equally to the humanities and the sciences, I was fascinated by geographic phenomena and their causes, reading Chinese National Geography and habitually analyzing problems from multiple angles. That habit matured into the principle that has guided all of my work since — first clarify the center of a problem, then think across its dimensions — and it eventually led me to the conviction that now defines my research: the structure of relationships encodes the essential information, and structure-aware learning is the most principled way to extract it.

I entered the Jianxing Honors College at Zhejiang University of Technology, chose a science-and-engineering track, and transferred into Internet of Things Engineering, contacting my first advisor in my sophomore year to begin research alongside my coursework. My undergraduate work centered on spatiotemporal data mining: I explored hydrological anomaly detection by combining hypergraph networks with reaction–diffusion equations, and studied deep-learning traffic-flow imputation with RNNs and GNNs. These projects taught me that physical equations and data-driven graphs are complementary lenses on the same spatiotemporal phenomena, and that nearby observations are deeply correlated through space-time regularities. Two experiences proved formative: as first student author, I built an edge–cloud collaborative aquaculture monitoring system — the first to apply SlowFast to fish-behavior detection (97.06% accuracy) — which was deployed in intensive American-shad farming and later accepted at IEEE SMC 2023; and I co-authored“Deep Learning for Time Series Forecasting: A Survey”(IJMLC 2025), now cited 428 times on Google Scholar, which taught me to synthesize a fragmented field into a clear map for newcomers.

This intuition matured into a systematic study of graph neural networks across three fronts. In multimodal recommendation, I treat the item graph itself as the object of learning — denoising noisy semantic graphs through behavior-conditioned diffusion (IGDMRec, IEEE TMM), mining co-semantic subgraphs with self-supervised hierarchical contrast (MMCSA), and reconciling frozen MLLM semantics with collaborative ranking through separate scoring paths (CS-SHP, WSDM 2027 under review). In compact GNN adaptation, I study what truly matters when a model must learn with only a handful of trainable parameters: the direction of weight updates, not merely their number (MapPerturb, ICLR 2027 under review). And in fairness, I formalize lower-tail risk management in influence maximization to protect disadvantaged groups from severe coverage failures (Information Sciences, under review).

Beyond model-centric questions, I extend this network perspective to problems with direct social consequence. During a remote internship at Clemson University’s Infrastructure, AI and Urban Nexus Lab, I quantified how a city’s heat accumulation is shaped by the structure of its street network. This grew into the Urban Thermal Resilience Index (UTRI), a

network-science metric of a street network's heat-ventilation potential. Linking UTRI to socioeconomic data in Washington, D.C., I found that thermal resilience correlates strongly with median household income — quantitative evidence that heat risk is an environmental-justice issue embedded in street-network planning.

These experiences converge on the direction I want to pursue in my PhD. Methodologically, I aim to build on graph neural networks and network science more broadly, using the lens of graph-structure information to uncover the latent representations behind observed phenomena and to approach concrete scientific questions through this structural view. In application, I want to direct these methods at urban climate resilience and social and natural environmental vulnerability — mining the phenomena and regularities that connect climate stress, environmental hazards, and social disadvantage through network structure, and designing purpose-built models that capture them and, ultimately, help reduce climate-driven inequity.

What I seek in a PhD is an environment where I can work alongside excellent, direction-matched supervisors and collaborators on questions that are challenging, deep, and worth answering — and where research value can be translated into social value. My starting point for choosing a program and a supervisor is this shared ambition: to do work that is rigorous and meaningful rather than merely incremental. I value settings that combine a serious academic culture with the quiet space for sustained, concentrated scholarship, and I look forward to bringing my graph-based methods to a group whose questions I can genuinely help answer.

My goal is to work on the front line of climate adaptation and disaster response, turning research into tools that help vulnerable communities. I bring to this goal a record of carrying work from idea to publication to deployment, and a conviction that good research is rigorous, grounded, and ultimately useful. A good beginning is common, but a good ending is rare — I value persistence above all. I hope to carry this thread into climate-change research, and to spend the PhD and beyond doing work that connects structure, society, and a warming world.